HEMODIALYSIS

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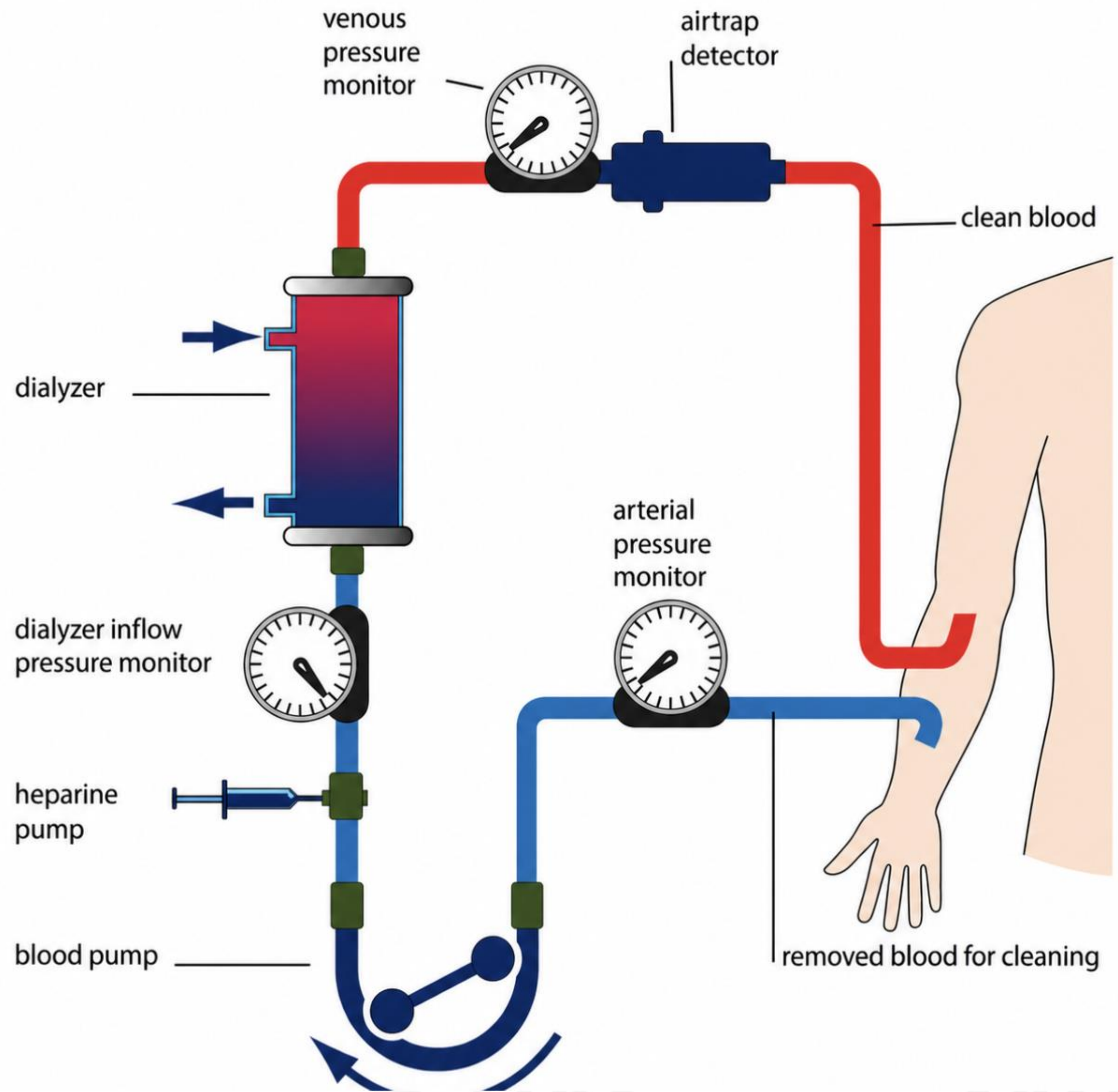
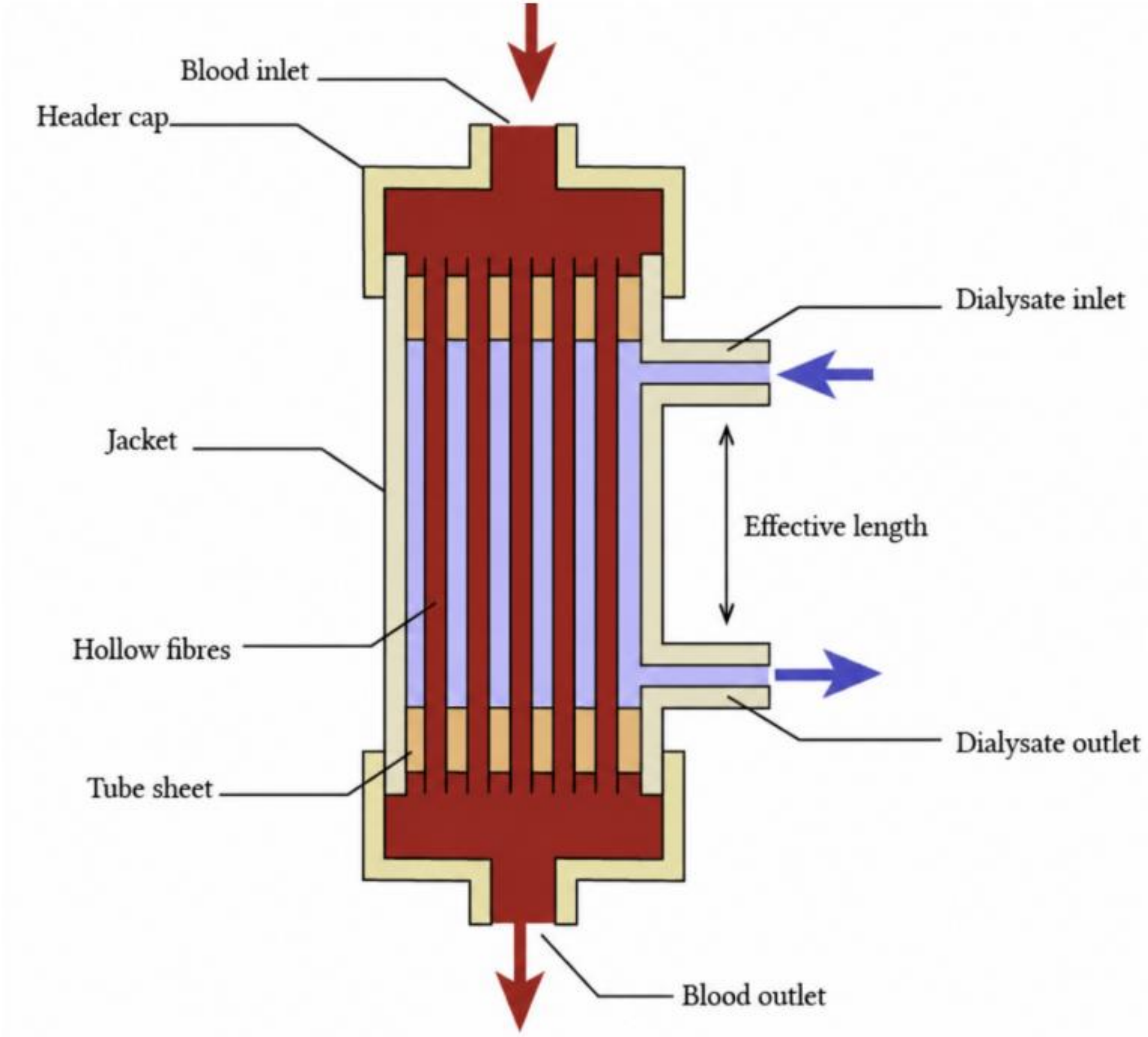
Objective

- Design a hemodialysis system for urea removal using mass transfer principles
- Calculate key transport parameters (diffusivity, mass transfer coefficient)
- Determine membrane requirements (surface area and number of fibers)
- Estimate system cost and evaluate performance factors



System Description

- Blood flows inside hollow fiber membranes
- Dialysate flows outside fibers in counter-current direction
- Semi-permeable membrane separates the two streams
- Urea diffuses from blood to dialysate due to concentration gradient
- Hollow fiber design provides high surface area for mass transfer



Governing Equations

- Overall urea mass balance:

$$Q_b\left(C_{b,i} - C_{b,o}\right) = Q_d\left(C_{d,o} - C_{d,i}\right)$$

- Temperature-corrected diffusivity:

$$D_{AB,T} = D_{AB,ref}\left(\frac{T}{T_{ref}}\right)\left(\frac{\mu_{ref}}{\mu_T}\right)$$

- Effective membrane diffusivity:

$$D_{Ae} = \varepsilon D_{AB,T}$$

- Overall mass transfer coefficient:

$$K_o \approx \frac{D_{Ae}}{\delta_m}$$

- Required membrane area:

$$A_m = \frac{Q_b\left(C_{b,i} - C_{b,o}\right)}{K_o \Delta C_{lm}}$$

Assumptions

- Steady-state operation (no accumulation of urea)
- Dialysate inlet concentration is negligible ($C_{d,i} \approx 0$)
- Membrane resistance dominates overall mass transfer
- Blood and dialysate flow rates are constant
- Uniform flow distribution through all fibers
- Physical properties are constant (density, viscosity, diffusivity)
- Membrane resistance dominates ($K_o \approx D_{Ae}/\delta_m$)
- Urea transport occurs only by diffusion

Diffusivity

- Urea diffusivity corrected for temperature at 37.4°C:

$$D_{AB,T} = 1.87 \times 10^{-9}\ m^2/s$$

- Temperature correction accounts for:
  - Change in fluid viscosity
  - Operating temperature difference
- Effective diffusivity in membrane:

$$D_{Ae} = \varepsilon D_{AB,T}$$

- Using membrane porosity ($\varepsilon = 0.20$):

$$D_{Ae} = 3.75 \times 10^{-10}\ m^2/s$$

Mass Transfer

- Overall coefficient was based on membrane resistance:

$$K_o \approx \frac{D_{Ae}}{\delta_m}$$

- Calculated overall mass transfer coefficient:

$$K_o = 1.50 \times 10^{-5}\ m/s$$

- Counter-current concentration differences:

$$\Delta C_1 = 1416.7\ mg/L$$
$$\Delta C_2 = 600\ mg/L$$

Log-mean concentration difference:

$$\Delta C_{lm} \approx 951\ mg/L$$

Membrane Area

- Urea removal rate from mass balance:

$$\dot{m}_A = Q_b\left(C_{b,i} - C_{b,o}\right)$$

- Design equation for membrane area:

$$A_m = \frac{Q_b\left(C_{b,i} - C_{b,o}\right)}{K_o \Delta C_{lm}}$$

- Using calculated values:

$$Q_b\left(C_{b,i} - C_{b,o}\right) = 1400\ mg/min$$

- Required membrane surface area:

$$A_m = 1.64\ m^2$$

- Larger membrane area increases urea removal efficiency

Fiber Design

- Membrane area related to hollow fiber geometry:

$$A_m = N_f \pi d_f L$$

- Solving for number of fibers:

$$N_f = \frac{A_m}{\pi d_f L}$$

- Using fiber diameter and Length :

$$d_f = 180\ \mu m \text{ and } L = 0.20\ m$$

- Calculated number of fibers:

$$N_f \approx 1.45 \times 10^4$$

Membrane Cost

- Membrane cost estimated using cost per unit area:

$$\text{Cost} = A_m \times \left(\frac{\$}{m^2}\right)$$

- Using literature value:

$$15\ \$/m^2$$

- Using calculated membrane area:

$$A_m = 1.64\ m^2$$

- Total membrane cost:

$$\text{Cost} \approx \$24.60$$

- Cost may increase when accounting for full system and manufacturing expenses

Improvements

- Increase membrane surface area to enhance urea removal
- Increase dialysate flow rate to maintain higher concentration gradient
- Increase membrane porosity to improve effective diffusivity
- Optimize fiber geometry to maximize mass transfer efficiency
- Ensure membrane selectivity to prevent loss of essential proteins

Conclusion